FYS5419 Project 1: Variational Quantum Eigensolvers and the Lipkin Model

Your Name

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Abstract

TODO: Write a concise summary of the problem, methods, main numerical results, and conclusions.

Contents

1 Introduction

TODO: Describe the goal of the project: gates, measurements, VQE, entanglement, and the Lipkin model.

The project studies small quantum systems using both classical diagonalization and variational quantum eigensolver (VQE) methods. The central numerical theme is to compare exact spectra with variational estimates and to interpret changes in eigenstates and entanglement as interaction strengths are varied.

2 Theory and numerical methods

TODO: Summarize the variational principle, VQE, Pauli decompositions, density matrices, partial traces, and von Neumann entropy.

For a normalized trial state $|\psi(\boldsymbol{\theta})\rangle$, the variational energy is

$$E(\boldsymbol{\theta}) = \langle \psi(\boldsymbol{\theta}) | H | \psi(\boldsymbol{\theta}) \rangle, \tag{1}$$

which is an upper bound to the ground-state energy when the trial state belongs to the Hilbert space of the Hamiltonian.

3 Part a: gates, Bell states, measurements, and entropy

TODO: Insert implementation details, tables of gate actions, measurement statistics, density matrices, and entropy results.

4 Part b: exact solution of the one-qubit Hamiltonian

TODO: Insert eigenvalue plots, eigenvector-composition plots, and interpretation of the state-character interchange.

5 Part c: one-qubit VQE

TODO: Describe the ansatz, optimizer, expectation-value evaluation, and comparison with exact diagonalization.

6 Part d: exact solution of the two-qubit Hamiltonian and entanglement

TODO: Insert spectra, reduced density matrices, von Neumann entropy, and discussion of level crossings.

7 Part e: two-qubit VQE

TODO: Describe the two-qubit ansatz, optimization results, and comparison with exact diagonalization.

8 Part f: Lipkin model and classical diagonalization

TODO: Show the Pauli decompositions, classical spectra for $J = 1$ and $J = 2$, and discussion of interaction-strength dependence.

9 Part g: Lipkin model with VQE

TODO: Describe the qubit encoding, ansatz, VQE results, and comparison with exact spectra.

10 Discussion

TODO: Discuss numerical reliability, optimizer convergence, ansatz expressivity, shot noise if used, and physical interpretation.

11 Conclusion

TODO: Summarize what was learned and give concise conclusions.

A Use of AI/LLM tools

TODO: Fill in this appendix if any AI/LLM tools were used. Delete or mark as not applicable only if no such tools were used.

A.1 Tools used

- Tool:

  TODO: tool name, version/access date, access method

A.2 Text contributions

Section	Level	Notes
Abstract	0	–
Introduction	0	–
Theory/methods	0	–
Results	0	–
Discussion/conclusion	0	–

A.3 Code contributions

File or notebook	Level	Description
`src/fys5419_project1/...`	0	–

A.4 Verification statement

All code, equations, and numerical results included in the submitted report have been checked and are understood by the author(s).

References

[1] CompPhysics, *Quantum Computing and Quantum Machine Learning*, course repository. https://github.com/CompPhysics/QuantumComputingMachineLearning

[2] M. Q. Hlatshwayo, Y. Zhang, H. Wibowo, R. LaRose, D. Lacroix, and E. Litvinova, "Simulating excited states of the Lipkin model on a quantum computer," *Physical Review C* **106**, 024319 (2022).
https://doi.org/10.1103/PhysRevC.106.024319